

Definition: A random variable is a function $X: S \rightarrow R$ which maps an outcome s in the sample space S to a number x in the range space R where $R \subseteq \mathbb{R}$. More simply, $X(s) = x$

Definition: A random variable is **discrete** if its range space is a countable set. A random variable is **continuous** if its range space is an uncountable set

Definition: For a discrete random variable X with range space R , the **probability mass function (pmf)** is the function

$$p(x) = \begin{cases} P(X=x), & \forall x \in R \\ 0, & x \notin R \end{cases}$$

$$0 \leq p(x) \leq 1 \text{ and } \sum_{x \in R} p(x) = 1$$

Definition: The process of dividing a probability mass function $p(x)$ by its sum to turn it into a pmf is called **normalisation** or **normalising** $p(x)$

Definition: The **cumulative distribution function (cdf)** $F(x)$ of a random variable X is the function that calculates the probability that X will not exceed a given value x :

$$F(x) = P(X \leq x)$$

$$\lim_{x \rightarrow -\infty} F(x) = 0, \quad \lim_{x \rightarrow +\infty} F(x) = 1, \quad F(x) \text{ is non-decreasing}$$

Theorem: Let the random variable X have cdf $F(x) = P(X \leq x)$ and range space R .

$$\text{Then } P(X > x) = 1 - F(x)$$

Theorem: Let the random variable X have cdf $F(x)$ and range space R .

$$\text{Then } P(X \in (a, b]) = F(b) - F(a)$$

Theorem: Let $x_0 \in R$ and $x \in R$ with $x_0 < x$ be two adjacent values of the range space of a discrete random variable.

(In case x is the smallest value of the range space, define x_0 to be some value such that $F(x_0) = 0$) The pmf of X is given in terms of its cdf by

$$p(x) = F(x) - F(x_0)$$